

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	ME-STE
Program Name	Master of Engineering in Structural Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Matthew Gombeda, Mehdi Modares
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	2
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Engineering in Structural Engineering program provides students with the knowledge needed to design the built environment. Students learn how built infrastructures (e.g., buildings, bridges, tunnels, dams, etc.) may be designed to resist the forces imposed upon them by external loads, gravity, wind, and earthquakes. Modern computer-aided design techniques and the latest national building codes dealing with structural steel, reinforced concrete, prestressed concrete, and masonry structures are utilized in course contents.

The primary learning objectives for this program are for students to:

1. Develop advanced technical knowledge of the fundamentals of structural engineering.
2. Demonstrate the capability to apply fundamental knowledge to structural engineering problems while utilizing underlying concepts of behavior, mechanics and design.

Artifact(s) used to evaluate student learning: Performance on assignments and exams in one of the required courses in the program. Performance in CAE 518 Advanced Reinforced Concrete is used in this cycle, as this course represents each learning goal and the depth and breadth of the program, with both structural analysis and structural design components, with a focus on reinforced concrete members.

Semester(s) in which artifacts are collected and assessed: End of each academic year

Sample size: Each student in the program who took the selected course

Data to report: Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator

Results

A total of 24 students were enrolled in the program in Fall 2024, including 9 who were pursuing the degree as a traditional master's student and 15 who were pursuing it as a co-terminal student. Among these 24 students, 8 took CAE 518 in this academic year (Spring 2025) and are thus used for program assessment in this cycle.

The following artifacts were used to assess performance toward each LO:

- LO #1 (Develop advanced technical knowledge of the fundamentals of structural engineering):
 - Performance on homework assignments 1-4, 8, and midterm exam, covering the following topics:
 - Derivation/premise of conventional design assumptions (e.g., Whitney stress block, strain compatibility, etc.), asymmetric cross-sections, mechanics of development length and anchorage, moment-curvature response and analysis methods, non-linear deflection mechanics and analysis methods, advanced behavior/mechanics of concrete columns in combined axial force and uniaxial moment
- LO #2 (Demonstrate the capability to apply fundamental knowledge to structural engineering problems while utilizing underlying concepts of behavior, mechanics and design):
 - Performance on homework assignments 5-7, 9, and final exam, covering the following topics:
 - Fundamental principles of two-way slabs, yield line theory/analysis, advanced mechanics of shear, deep beam theory, strut and tie methods, torsion, columns with biaxial bending, and slender columns

Table 1 shows results from the 8 students in the program across LOs #1 and #2. Scores range from 0 to 100 on a fractional basis. Definitions of proficiency levels are as follows: ≥ 0.9 (proficiency), 0.7-0.9 (meets expectations), < 0.7 (does not meet expectations). Distributions of proficiency levels are shown in Figure 1. Artifact samples are archived in Canvas.

Table 1. Results from evaluation rubric applied to LOs #1 and #2 for this academic year

student	LO1	LO1 - DMP	LO2	LO2 - DMP
student 1	0.92	P	0.94	P
student 2	0.92	P	0.92	P
student 3	0.64	D	0.56	D
student 4	0.91	P	0.94	P
student 5	0.91	P	0.92	P
student 6	0.91	P	0.90	P
student 7	0.57	D	0.55	D
student 8	0.91	P	0.95	P

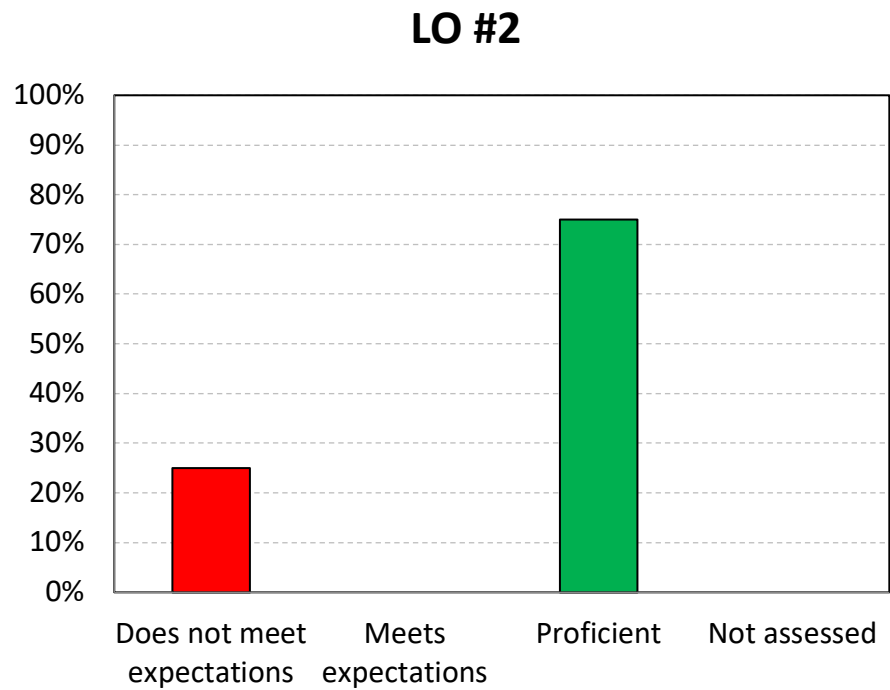
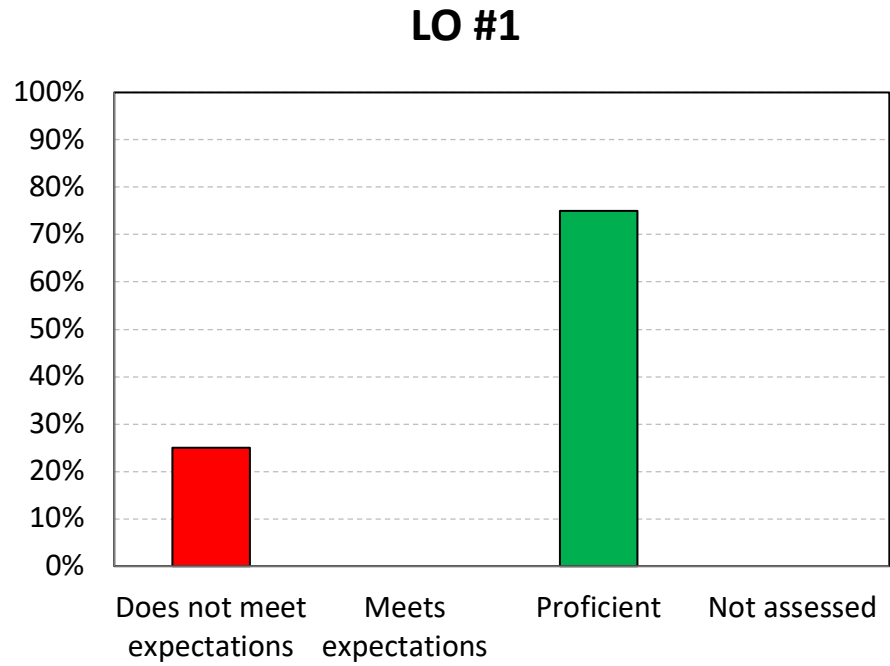


Figure 1. Distribution of proficiency levels for LOs #1 and #2for this academic year

Description of findings

Key findings from this academic year and cohort include:

- Only 2 out of 8 students sampled (25%) did not meet or exceed expectations for both LO #1 and LO #2, suggesting a high degree of proficiency in meeting learning objectives in this cycle.
 - One of the two “Does not meet expectations” students fell behind in homework assignments due to a personal issue, which the instructor does not believe is a true indication of the student’s learning potential nor does it signal a weakness in the course objectives or content.
- Overall, a high level of proficiency demonstrated in this course raised no major concerns in this cycle.

Prior CI plan impact

In the previous assessment report, institutional reviewers made several recommendations to improve upon our assessment plan, including converting to an updated format, indicating more clearly which artifacts are used and collected, and to improve upon our CI plan. None of the performance shown above raises major concerns for background preparation for the program or ability to execute in the program.

This cycle’s CI plan

In this cycle, we revised our learning objectives to 2 simply stated objectives, crafting language that allows for a broader characterization of the primary goals of the program. We also articulated a clearer plan for assessing the program using more than one course if needed in any given year. In this cycle, we selected CAE 518 for assessment since its assignments 1-4 and 8, and midterm exam are well-aligned with preparation for LO #1 while the final exams and assignments 5, 6, 7 and 9 are well-aligned with preparation for LO #2.